T. A. MARRYSHOW COMMUNITY COLLEGE

DEPARTMENT OF INFORMATION TECHNOLOGY

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COURSE NAME AND NUMBER: ADVANCE PROGRAMMING CIT244

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SCOPE

Project Title: Password Strength Analyser

The Password Strength Analyser is a Java desktop application that allows users to input a password and receive instant feedback on how strong or weak it is. The application evaluates the password based on criteria such as length, use of uppercase and lowercase letters, inclusion of numbers, and special characters. It will display a strength rating, Weak, Moderate, Strong, or Very Strong, along with a visual indicator and suggestions for improvement. The software is designed to be simple, user-friendly, and accessible to users with no technical background. It will be built as a GUI form application using Java Swing in IntelliJ IDEA.

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REASON FOR YOUR PROJECT

Weak passwords remain one of the leading causes of account breaches and identity theft in today's digital world. Many users do not know what makes a password secure until after they have already been compromised. The purpose of this project is to provide a real-time, easy-to-use tool that guides users toward creating stronger passwords at the point of creation, before any damage is done. Rather than learning about password security after a breach, users will receive immediate, actionable guidance. This project also demonstrates the practical application of Java GUI programming, event handling, and string manipulation, all core concepts in Advanced Programming.

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SOFTWARE TO BE USED

The application will be developed using the following:

- **Programming Language:** Java
- **IDE:** IntelliJ IDEA
- **GUI Framework:** Java Swing (JFrame form application)
- **Components Used:** JTextField, JPasswordField, JLabel, JButton, JProgressBar, and color-coded feedback panels
- **No external database is required**, as all analysis logic runs locally within the application at runtime

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ALGORITHM AND FLOWCHART OF THE SOFTWARE

Step 1 - Start: The application launches and displays the GUI to the user.

Step 2 - User Input: The user types a password into the password input field.

Step 3 - Trigger Analysis: The analysis is triggered either in real time as the user types (using a document listener) or when the user clicks the "Analyse" button.

Step 4 - Length Check: The algorithm checks whether the password is at least 8 characters long. If true, 1 point is added to the score.

Step 5 - Uppercase Check: The algorithm checks whether the password contains at least one uppercase letter (A–Z). If true, 1 point is added.

Step 6 - Lowercase Check: The algorithm checks whether the password contains at least one lowercase letter (a–z). If true, 1 point is added.

Step 7 - Digit Check: The algorithm checks whether the password contains at least one numeric digit (0–9). If true, 1 point is added.

Step 8 - Special Character Check: The algorithm checks whether the password contains at least one special character such as !, @, #, $, %, ^, &, or *. If true, 1 point is added.

Step 9 - Calculate Score: The total score is calculated out of 5 based on how many criteria passed.

Step 10 - Determine Strength Rating:

- Score 0 - 1 → **Weak**
- Score 2 - 3 → **Moderate**
- Score 4 → **Strong**
- Score 5 → **Very Strong**

Step 11 - Display Results: The GUI updates to show the strength label, a color-coded progress bar (red → orange → yellow → green), and a list of suggestions for any criteria the password did not meet.

Step 12 - End: The user may clear the field and test another password or close the application.

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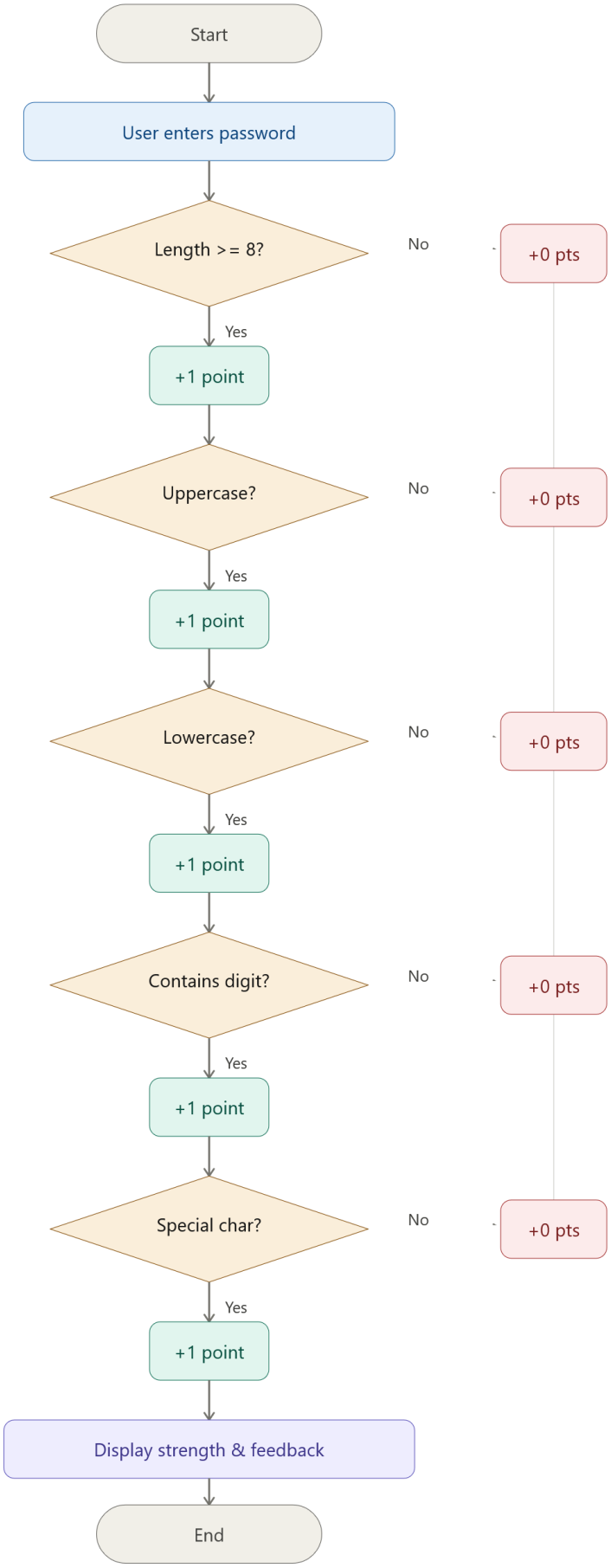

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